FRACTIONS

A ***fraction*** shows a part of a whole. For example, if Joe cuts a pizza into 8 equal slices, then eats 3 of the 8 slices, he ate $\frac{3}{8}$ of the pizza, and $\frac{5}{8}$ of the pizza is left.

The number at the top of a fraction is called the ***numerator***. The numerator refers to the number of parts out of the total number of parts making up the whole. For example, in the fraction $\frac{3}{8}$, 3 is the numerator. The number at the bottom of the fraction is called the ***denominator***. It tells you the total number of parts making up the whole. For example, in the fraction $\frac{3}{8}$, 8 is the denominator

In order to work with fractions, you must first know the differences between types of fractions. Fractions may exist in a number of forms. A ***proper***, or ***common***, fraction is one that cannot be reduced to lower numbers, and the numerator (top number) is smaller than the denominator (bottom number) (e.g., $\frac{3}{4}$). An ***improper*** fraction can be reduced to a mixed number because the numerator is larger than the denominator (e.g., $\frac{11}{2}$ can change to $5\frac{1}{2}$). A ***mixed number*** is the combination of a whole number and a proper fraction (e.g., $5\frac{1}{2}$). An ***unreduced fraction*** is a fraction that has not been reduced to its lowest terms (e.g., $\frac{2}{4}$ can be reduced to $\frac{1}{2}$).

For the sample problems below, please select the answer that best represents each of the following types of fractions. Mark the corresponding letter in the Practice Answer Box.

✎ PRACTICE: Types of Fractions

1. A proper fraction	(A) $\frac{53}{8}$	(B) $\frac{7}{8}$	(C) $\frac{7}{5}$	(D) $\frac{16}{16}$
2. An improper fraction	(A) $7\frac{1}{2}$	(B) $\frac{8}{5}$	(C) $5\frac{2}{3}$	(D) $\frac{6}{9}$
3. A mixed number	(A) $13\frac{4}{5}$	(B) $\frac{35}{67}$	(C) $\frac{7}{16}$	(D) $\frac{7}{7}$
4. An unreduced fraction	(A) $\frac{4}{8}$	(B) $\frac{1}{3}$	(C) $\frac{13}{14}$	(D) $5\frac{1}{2}$

Practice Answer Box:

PRACTICE				
1.	(A)	(B)	(C)	(D)
2.	(A)	(B)	(C)	(D)
3.	(A)	(B)	(C)	(D)
4.	(A)	(B)	(C)	(D)

In order to work with fractions, you need to know how to:

A. Reduce Fractions

B. Find A Lowest Common Denominator

C. Change Improper Fractions To Whole Or Mixed Numbers

D. Change Mixed Numbers To Improper Fractions

Then you will be ready to learn:

E. Adding Fractions

F. Subtracting Fractions

G. Multiplying Fractions

H. Dividing Fractions

A. Reducing Fractions

Reducing a fraction means dividing the numerator and the denominator by a number that goes into both of them evenly. When you reduce a fraction, you change the numbers in the fraction, but not the value of the fraction. For example: $\frac{1}{2}$ is the same amount as $\frac{4}{8}$ or $\frac{2}{4}$. The numbers in the fraction change, but the value of the fraction stays the same.

☞ EXAMPLE: $\frac{4}{8} \div \frac{4}{4} = \frac{1}{2}$

Step 1: Divide 4 and 8 by a number that goes evenly into both of them. 2 divides evenly into both 4 and 8. $\frac{4}{8} \div \frac{2}{2} = \frac{2}{4}$

Step 2: Check to see if any other number besides 1 goes evenly into both 1 and 2. If so, reduce the fraction again. $\frac{2}{4} \div \frac{2}{2} = \frac{1}{2}$

Step 3: Check to see if any other number besides 1 goes evenly into both 1 and 2. No other number does, so $\frac{1}{2}$ is reduced as far as it can go. (A fraction is in LOWEST TERMS when it is reduced as far as it can go).

Now try the following practice problems.

✎ PRACTICE: Reducing Fractions

(1) $\frac{3}{9} =$ (2) $\frac{4}{8} =$ (3) $\frac{6}{12} =$ (4) $\frac{4}{2} =$ (5) $\frac{9}{12} =$

(6) $\frac{6}{10} =$ (7) $\frac{5}{10} =$ (8) $\frac{12}{14} =$ (9) $\frac{8}{12} =$ (10) $\frac{20}{30} =$

B. Finding A Lowest Common Denominator

The *lowest common denominator* is the smallest possible number that can be divided evenly by the denominators of all the fractions in an equation. To find a lowest common denominator, you must decide what number each denominator can be divided into evenly. If this is too difficult, simply multiply both the numerator and denominator of each fraction by the respective denominator of every other fraction.

☞ EXAMPLE: $\frac{3}{4}$ and $\frac{2}{3}$; lowest common denominator = 12

Step 1: Think of the lowest number that both denominators (4 and 3) divide into evenly. 12 divided by either 4 or 3 results in a whole number.

Step 2: If unable to do Step 1, try multiplying the numerators and denominators of each fraction by the denominators of every other fraction in the equation.

$\frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$
$\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}$

When you have to make all the denominators in a problem the same, divide the biggest denominator by the smaller one.

☞ EXAMPLE: $\frac{3}{4}$ and $\frac{13}{32}$

Step 1: Divide the large denominator by the small one.

$32 \div 4 = 8$

Step 2: Multiply both the numerator and denominator of the first equation by the result of Step 1. The result is an equivalent fraction of $\frac{3}{4}$.

Now both fractions have a common denominator of 32.

$\frac{3}{4} \times \frac{8}{8} = \frac{24}{32}$

Another way to find a common denominator is to look at multiples of the larger denominator, going up until you find one that all the denominators divide into evenly.

☞ EXAMPLE: $\frac{1}{6} + \frac{3}{8}$

Step 1: Think of the multiples of the largest number (8).

8, 16, 24, 32, ?

Step 2: Try to divide the lower number into the multiples of the higher number.

(Will 6 go into 8? No. Into 16? No. Into 24? Yes!)

B. Finding A Lowest Common Denominator (cont.)

Now try the following practice problems.

✎ PRACTICE: Finding a Lowest Common Denominator

(1) $\frac{4}{5} + \frac{9}{10}$ Lowest Common Denominator = _____

(2) $\frac{1}{6} + \frac{5}{8} - \frac{4}{7}$ Lowest Common Denominator = _____

(3) $\frac{1}{3} - \frac{5}{7}$ Lowest Common Denominator = _____

(4) $\frac{4}{9} - \frac{5}{6} + \frac{1}{4}$ Lowest Common Denominator = _____

(5) $\frac{3}{16} + \frac{5}{8} + \frac{1}{2}$ Lowest Common Denominator = _____

C. Changing Improper Fractions To Whole Or Mixed Numbers

In some cases, you may wish to change an improper fraction to a whole number or a mixed number. This is done by dividing the numerator by the denominator and then creating a mixed fraction by using the remainder as the numerator and keeping the same denominator. If there is no remainder, you have a whole number.

☞ EXAMPLE: $\frac{16}{3} = 5\frac{1}{3}$

Step 1: Divide the bottom (denominator) into the top (numerator).

$$3\overline{)\,16}\quad 5,\ -15,\ 1 \leftarrow \text{remainder}$$

Step 2: Write the remainder over the original bottom number (denominator) and place to the right of the whole number $5\frac{1}{3}$

Step 3: Reduce the fraction, if necessary.

Now try the following practice problems.

✎ PRACTICE: Changing improper fractions to whole or mixed numbers

(1) $\frac{28}{8} =$ (2) $\frac{14}{3} =$ (3) $\frac{53}{20} =$ (4) $\frac{9}{9} =$ (5) $\frac{16}{6} =$

D. Changing Mixed Numbers To Improper Fractions

Similarly, changing mixed numbers into improper fractions may be important when dividing or multiplying fractions.

☞ EXAMPLE: $2\frac{3}{4} = \frac{11}{4}$

Step 1:	Multiply the bottom number (denominator) by the whole number.	4 x 2 = 8
Step 2:	Add the top number (numerator).	8 + 3 = 11
Step 3:	Place the total over the bottom number.	$\frac{11}{4}$

Now try the following practice problems.

✎ PRACTICE: Changing mixed numbers to improper fractions

(1) $5\frac{2}{11} =$ (2) $7\frac{5}{12} =$ (3) $4\frac{3}{7} =$ (4) $10\frac{2}{3} =$ (5) $9\frac{7}{8} =$

E. Adding Fractions

1. Adding fractions with the same denominator:

When adding fractions with the same denominator, add only the numerators. The bottom numbers (denominator) stay the same.

☞ EXAMPLE: $\frac{3}{5} + \frac{1}{5} =$ $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$

Step 1: Add the numerators. $3 + 1 = 4$

Step 2: The denominator stays the same. $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$

Step 3: If necessary, change the total to a mixed number (see pg. 17 "Changing Improper Fractions To Whole Or Mixed Numbers"), or reduce.

Now try the following practice problems.

✎ PRACTICE: Adding fractions

(1) $\frac{2}{7} + \frac{1}{7} =$

(2) $\frac{1}{3} + \frac{1}{3} =$

(3) $\frac{4}{14} + \frac{2}{14} =$

(4) $\frac{2}{9} + \frac{4}{9} =$

(5) $\frac{3}{13} + \frac{8}{13} =$

E. Adding Fractions (cont.)

2. Adding fractions with different denominators:

When adding fractions with different denominators, find a lowest common denominator, change the fractions so that they have the same (common) denominator, and then add the numerators.

☞ EXAMPLE: $\frac{3}{5} + \frac{4}{15} = \frac{13}{15}$

Step 1:	Find a common denominator (see pg. 15 "Finding A Lowest Common Denominator").	Since 5 divides evenly into 15, 15 is the lowest common denominator.
Step 2:	Multiply both the numerator and the denominator so that each denominator equals the lowest common denominator.	3 (× 3) = 9 / 5 (× 3) = 15 = $\frac{9}{15}$
Step 3:	Add the numerators.	$\frac{9}{15} + \frac{4}{15} = \frac{13}{15}$
Step 4:	If necessary, change the total to a mixed number (see pg. 17 "Changing Improper Fractions To Whole Or Mixed Numbers"), or reduce.	

Now try the following practice problems.

✎ PRACTICE: Adding fractions with different denominators

(1) $\frac{3}{5} + \frac{8}{15} + \frac{2}{3} =$

(2) $\frac{3}{5} + \frac{1}{2} + \frac{7}{10} =$

(3) $6\frac{5}{48} + 2\frac{5}{8} =$

(4) $\frac{5}{9} + \frac{11}{36} =$

(5) $\frac{5}{8} + \frac{7}{32} + \frac{9}{16} =$

F. Subtracting Fractions

1. Subtracting fractions with different denominators:

When subtracting fractions with different denominators, find the lowest common denominator, change the fractions so that they have the same (common) denominator, then subtract the second fraction from the first.

☞ EXAMPLE: $\frac{1}{2} - \frac{3}{8} = \frac{1}{8}$

Step 1:	Find the lowest common denominator (2 and 8 both divide evenly into 8).	$8 \div 8 = 1$ $8 \div 2 = 4$
Step 2:	Multiply both the numerator and the denominator so that each denominator equals the lowest common denominator.	$\frac{1}{2} = \frac{4}{8}$ ($\frac{1}{2} \times \frac{4}{4} = \frac{4}{8}$)
Step 3:	Subtract the fractions.	$\frac{4}{8} - \frac{3}{8} = \frac{1}{8}$
Step 4:	Reduce, if necessary.	

Now try the following practice problems.

✎ PRACTICE: Subtracting fractions

(1) $\frac{2}{3} - \frac{1}{4} =$

(2) $\frac{9}{16} - \frac{1}{4} =$

(3) $\frac{7}{8} - \frac{5}{6} =$

(4) $8\frac{5}{9} - 3\frac{1}{6} =$

(5) $\frac{2}{3} - \frac{2}{9} =$

F. Subtracting Fractions (cont.)

2. Borrowing and subtracting fractions:

When subtracting fractions from whole numbers, change the whole number to a mixed number, and then subtract the proper fraction from the fraction in the mixed number.

☞ EXAMPLE: $9 - \frac{3}{5} = 8\frac{2}{5}$

Step 1:	Change the whole number to a mixed number. Borrow 1 from the whole number and put it in the form of a fraction with both the numerator and denominator equal to the denominator of the proper fraction.	$9 = 8\frac{5}{5}$
Step 2:	Subtract the fractions from the fraction you created in Step 1. If subtracting a mixed number from a whole number, also remember to subtract the whole numbers.	$8\frac{5}{5} - \frac{3}{5} = 8\frac{2}{5}$
Step 3:	Reduce if necessary	

Now try the following practice problems.

✎ PRACTICE: Borrowing and subtracting fractions

(1) $7 - \frac{5}{8} =$

(2) $14 - 4\frac{3}{5} =$

(3) $10 - \frac{7}{9} =$

(4) $3 - \frac{3}{4} =$

(5) $2 - \frac{1}{4} =$

G. Multiplying Fractions

1. Multiplying fractions with fractions:

When multiplying fractions by other fractions, you simply multiply the numerators by the numerators and the denominators by the denominators and reduce, if needed.

☞ EXAMPLE: $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Step 1: Multiply the numerators.

Step 2: Multiply the denominators.

Step 3: Reduce when necessary.

$$\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$

Now try the following practice problems.

✎ PRACTICE: Multiplying fractions with fractions

(1) $\frac{2}{3} \times \frac{1}{4} =$

(2) $\frac{1}{3} \times \frac{1}{5} =$

(3) $\frac{1}{2} \times \frac{1}{4} \times \frac{3}{5} =$

(4) $\frac{1}{5} \times \frac{6}{7} =$

(5) $\frac{1}{2} \times \frac{3}{4} =$

G. Multiplying Fractions (cont.)

2. Multiplying whole numbers and fractions:

When multiplying whole numbers and fractions, you must first change the whole number into an improper fraction. Then, multiply the numerators by the numerators and the denominators by the denominators.

☞ EXAMPLE: $\frac{3}{5} \times 6 = 3\frac{3}{5}$

Step 1: Give the whole number a denominator of 1. $6 = \frac{6}{1}$

Step 2: Multiply the fractions. $\frac{3}{5} \times \frac{6}{1} = \frac{18}{5}$

Step 3: Reduce when possible. $\frac{18}{5} = 3\frac{3}{5}$

Now try the following practice problems.

✎ PRACTICE: Multiplying whole numbers and fractions

(1) $\frac{2}{9} \times 4 =$

(2) $5 \times \frac{2}{3} =$

(3) $\frac{1}{4} \times 5 =$

(4) $\frac{2}{7} \times 6 =$

(5) $8 \times \frac{2}{5} =$

G. Multiplying Fractions (cont.)

3. Multiplying mixed numbers and fractions:

When multiplying mixed numbers, you must change the mixed number to an improper fraction, then multiply the numerators by the numerators and the denominators by the denominators. You may then have to change the improper fraction back into a mixed number.

☞ EXAMPLE: $\frac{3}{4} \times 5\frac{1}{2} = 4\frac{1}{8}$

Step 1:	Change the mixed number to an improper fraction (see pg. 18 "Changing Mixed Numbers To Improper Fractions"). Multiply the denominator by the whole number and add the numerator.	$2 \times 5 + 1 = 11$ $5\frac{1}{2} = \frac{11}{2}$
Step 2:	Multiply the fractions	$\frac{3}{4} \times \frac{11}{2} = \frac{33}{8}$
Step 3:	Reduce the fraction, if necessary.	
Step 4:	Change the improper fraction to a mixed number.	$4\frac{1}{8}$ $8\overline{)33}$ $-\ 32$ 1

Now try the following practice problems.

✎ PRACTICE: Multiplying mixed numbers and fractions

(1) $\frac{3}{4} \times 5\frac{1}{2} =$

(2) $\frac{5}{7} \times 2\frac{2}{3} =$

(3) $1\frac{1}{2} \times 2\frac{2}{3} =$

(4) $\frac{4}{5} \times 2\frac{1}{3} =$

(5) $3\frac{1}{3} \times \frac{4}{9} =$

H. Dividing Fractions

To divide a fraction, invert (turn upside-down) the fraction you are dividing by. Then multiply.

☞ EXAMPLE: $\frac{1}{2} \div \frac{1}{6} = 3$

Step 1:	Invert the fraction you are dividing by.	$\frac{1}{6}$ inverted is $\frac{6}{1}$
Step 2:	Multiply the numerators by the numerators and the denominators by the denominators.	$\frac{1}{2} \times \frac{6}{1} = \frac{6}{2}$
Step 3:	Reduce, if necessary.	$\frac{6}{2} = 3$

Sometimes you can CANCEL in division. You divide both the numerator of one fraction and the denominator of another by the lower of the two numbers or another whole number in order to reduce the size of the number to multiply. INVERT before you CANCEL, and MULTIPLY after you CANCEL.

☞ EXAMPLE: $\frac{2}{3} \div \frac{1}{6} = 4$

Step 1:	Invert the fraction you are dividing by.	$\frac{1}{6}$ inverted is $\frac{6}{1}$
Step 2:	Cancel.	$\frac{2}{3\ (\div 3)} \times \frac{6\ (\div 3)}{1} = \frac{2}{1} \times \frac{2}{1}$
Step 3:	Multiply the numerators by the numerators and the denominators by the denominators.	$\frac{2}{1} \times \frac{2}{1} = \frac{4}{1}$
Step 4:	Reduce, if necessary.	$\frac{4}{1} = 4$

Before dividing, change *any* MIXED NUMBERS into improper fractions.

☞ EXAMPLE: $1\frac{1}{4} \div \frac{2}{3} =$

Step 1:	Change the mixed number to an improper fraction.	$\frac{5}{4} \div \frac{2}{3} =$
Step 2:	Invert and multiply.	$\frac{5}{4} \times \frac{3}{2} = \frac{15}{8}$
Step 3:	Reduce your answer.	$\frac{15}{8} = 1\frac{7}{8}$

H. Dividing Fractions (cont.)

Now try the following practice problems.

✎ PRACTICE: Dividing fractions

(1) $\frac{1}{2} \div \frac{1}{4} =$

(2) $\frac{2}{3} \div 3 =$

(3) $\frac{1}{6} \div 2 =$

(4) $\frac{1}{4} \div \frac{1}{8} =$

(5) $1\frac{3}{4} \div \frac{1}{4} =$

DECIMALS

A. Adding Decimals

To add decimals, line up the numbers keeping the decimal points under each other.

☞ EXAMPLE: .28 + .9 + .327 = 1.507

Step 1: Set up the problem. Line up the numbers with the decimal points in one column.

```
  .28
  .9
+ .327
```

Step 2: Add each column. Make sure to bring the decimal point down in its column.

```
  .28
  .9
+ .327
 1.507
```

When you add whole numbers with decimals or mixed decimals, put a decimal point at the right of the whole number. Then line up the numbers with the decimal points under each other.

☞ EXAMPLE: 26 + 3.85 + .327 = 30.177

Step 1: Put a decimal point at the right of 26.

```
26.
```

Step 2: Set up the problem. Line the numbers up with the decimal point in one column.

```
 26.
  3.85
+  .327
```

Step 3: Add each column.

```
 26.
  3.85
+  .327
 30.177
```

Now try the following practice problems.

✎ PRACTICE: Adding decimals

(1) .3 + .07 + 5.3 =

(2) .75 + 7.50 + .075 =

(3) 6.007 + 5.3 + .75 =

(4) 3.5 + 75.39 =

(5) 59 + 5.4 + .07 =

B. Subtracting Decimals

To subtract decimals, line up the numbers with the decimal points under each other. Put a point at the right of each whole number. Put zeros at the right until each decimal has the same number of places (you will need the extra zeros for borrowing).

☞ EXAMPLE: 94 – 89.6 = 4.4

Step 1:	Put a decimal point at the right of 94.	94.
Step 2:	Set up the problem. Insert zeros as needed. Line the numbers up with the decimal points in the same columns.	94.0 – 89.6
Step 3:	Borrow, subtract, and bring down the decimal point.	94.0 – 89.6 4.4
	To borrow, if a digit (i.e., 4, 8, 0, etc.) in the upper number is smaller than the digit directly below in the lower number, than you may "borrow" (subtract) 1 from the digit to the left of the upper digit, and add ten to that upper digit.)	8 13 10 ~~9~~ ~~4~~ . 0 – 8 9 . 6 0 4 . 4

Now try the following practice problems.

✎ PRACTICE: Subtracting decimals

(1) 75 – 10.3 =

(2) 4.507 – 3.6 =

(3) 100.73 – 47.4 =

(4) 39.5 – 27.06 =

(5) 97.03 – 36.05 =

C. Multiplying Decimals

When you multiply decimals, you do not have to line up the decimal points. Just set the numbers up for easy multiplication. Multiply, then count how many decimal places were in the two numbers you multiplied. Put the same number of places in your answer.

☞ EXAMPLE: 2.54 × 6 = 15.24

Step 1:	Set up the problem.
Step 2:	Multiply.
Step 3:	Make sure your answer has the same number of decimal places as the two numbers in your problem.

	2.54	(two decimals)
×	6	(no decimals)
	15.24	(two decimals)

Now try the following practice problems.

✎ PRACTICE: Multiplying decimals

(1) 73.5 x 6 =

(2) 92.3 x 15.04 =

(3) 17.6 x 9.07 =

(4) 10 x 2.5 =

(5) 6.34 x 5.71 =

D. Dividing Decimals

1. Dividing decimals by whole numbers:

When you divide a decimal by a whole number, line up your problem carefully. Bring up the decimal point into the answer just above its position in the problem.

☞ EXAMPLE: 4.25 ÷ 5 = 0.85

Step 1: Set up the problem as you would for whole number division.

Step 2: Divide and bring the decimal point up into the answer, directly above its position in the problem.

```
   .85
5 ⟌4.25
  4 0
   25
   25
    0
```

Now try the following practice problems.

✎ PRACTICE: Dividing decimals by whole numbers

(1) 7.5 ÷ 3 =

(2) 75.6 ÷ 45 =

(3) 13.09 ÷ 7 =

(4) 47.4 ÷ 6 =

(5) 97.24 ÷ 2 =

D. Dividing Decimals (cont.)

2. Dividing decimals by decimals:

To divide by a decimal, first change the decimal to a whole number by moving the decimal point of the number outside the bracket to the right end. Then move the decimal point in the number being divided the same number of spaces to the right.

☞ EXAMPLE: $6.72 \div 1.6 = 4.2$

Step 1:	Set up the problem as you would to divide whole numbers.	$1.6\overline{)6.72}$
Step 2:	Change the number outside the bracket into a whole number by moving the decimal place as far to the right as it will go (the decimal disappears).	$1.6\overline{)6.72}$
Step 3:	Move the decimal point of the number inside the bracket the same number of spaces to the right.	$16\overline{)6.7.2}$
Step 4:	Bring the point up in the answer directly over its new position and divide.	$16\overline{)67.2}$ → 4.2; 64; 32; 32; 0

Now try the following practice problems.

✎ PRACTICE: Dividing decimals by decimals

(1) $.36 \div .6 =$

(2) $3.3 \div 1.2 =$

(3) $44.1 \div .7 =$

(4) $18.98 \div .26 =$

(5) $19.98 \div 5.4 =$

D. Dividing Decimals (cont.)

3. Dividing whole numbers by decimals

To divide a whole number by a decimal, first put a decimal point to the right of the whole number. Then move the decimal point of the number outside the bracket as far to the right as it will go. Then move the decimal point of the number inside the brackets the same number of spaces to the right (you may have to add zeros).

☞ EXAMPLE: $9 \div .25 = 36$

Step 1:	Set the problem up as you would to divide whole numbers. Put a decimal point at the right of the whole number.	$.25\overline{)9.}$
Step 2:	Change the outside number to a whole number by moving the decimal as far to the right as it will go.	$.25\overline{)9.}$
Step 3:	Move the decimal point of the number in the bracket the same number of spaces to the right, adding zeros to give enough places.	$25\overline{)9.00}$
Step 4:	Bring the point up in the answer directly above its new place in the number inside the bracket and divide.	$\begin{array}{r} 36. \\ 25\overline{)900.} \\ \underline{75} \\ 150 \\ \underline{150} \\ 0 \end{array}$

Now try the following practice problems.

✎ PRACTICE: Dividing whole numbers by decimals

(1) $9 \div .6 =$

(2) $37 \div .25 =$

(3) $343 \div .7 =$

(4) $1178 \div .019 =$

(5) $2592 \div .36 =$

✎ PRACTICE PROBLEMS: NUMERICAL COMPUTATION

How to do a Numerical Computation problem on the test:

Step 1: Write the problem on your scratch paper, making sure that all the decimal places line up when you are adding or subtracting.

Step 2: Do your arithmetic.

Step 3: Compare your answer with the possible answers on the test.

Step 4: If your answer is the same as one of the answers given on the test, blacken the letter of that answer sheet with your pencil.

Step 5: If none of the answers on the test match your answer, recheck your arithmetic.

Step 6: If you have completed all the problems and there is time remaining, recheck the problems you had the most trouble with.

1. $\frac{2}{5} + \frac{1}{5} =$	(A) $\frac{3}{10}$	(B) $\frac{4}{5}$	(C) $\frac{3}{5}$	(D) $\frac{1}{10}$	(E) $\frac{1}{5}$
2. .3 + .4 + .7 =	(A) 1.4	(B) 14	(C) 1.7	(D) .15	(E) 0
3. .9 – .2 =	(A) .1	(B) .7	(C) 7.8	(D) .6	(E) .11
4. $\frac{1}{3} + \frac{5}{6} =$	(A) $\frac{4}{3}$	(B) $\frac{4}{6}$	(C) $\frac{6}{18}$	(D) $\frac{7}{18}$	(E) $1\frac{1}{6}$
5. $\frac{1}{5} + \frac{2}{7} =$	(A) $\frac{17}{35}$	(B) $\frac{3}{12}$	(C) $\frac{2}{12}$	(D) $\frac{3}{35}$	(E) $\frac{3}{7}$
6. $\frac{5}{8} - \frac{3}{8} =$	(A) $\frac{8}{8}$	(B) $\frac{2}{16}$	(C) $\frac{1}{8}$	(D) $\frac{3}{4}$	(E) $\frac{1}{4}$
7. 6.34 × 9 =	(A) 5.706	(B) 57.07	(C) 5.77	(D) 57.06	(E) 50.6
8. 4.912 ÷ 16 =	(A) 3.07	(B) 3.08	(C) 0	(D) 30.7	(E) .307
9. $\frac{25}{37} - \frac{13}{37} =$	(A) $\frac{12}{35}$	(B) $\frac{12}{37}$	(C) $\frac{12}{25}$	(D) $\frac{38}{37}$	(E) $\frac{3}{7}$
10. 5.3 + 70.6 =	(A) 7.59	(B) 75.9	(C) 759	(D) .759	(E) 7
11. $9 \div 1\frac{1}{4} =$	(A) $7\frac{2}{5}$	(B) $7\frac{1}{5}$	(C) $6\frac{3}{5}$	(D) $7\frac{5}{36}$	(E) $\frac{9}{50}$
12. 1.65 – 1.49 =	(A) 16	(B) 1.06	(C) .15	(D) 0.16	(E) .11

PRACTICE PROBLEMS: NUMERICAL COMPUTATION(cont.)

13. $5 - 3\frac{1}{7} =$ (A) $2\frac{6}{7}$ (B) $\frac{17}{7}$ (C) $1\frac{6}{7}$ (D) $1\frac{5}{7}$ (E) $2\frac{13}{14}$

14. $5 \times \frac{4}{5} =$ (A) $5\frac{4}{5}$ (B) 4 (C) $5\frac{5}{20}$ (D) $20\frac{1}{5}$ (E) $5\frac{1}{5}$

15. $81.6 \times .7 =$ (A) 571.2 (B) 57.23 (C) 5.712 (D) 57.12 (E) 57

16. $\frac{2}{7} \times \frac{4}{3} =$ (A) $\frac{6}{28}$ (B) $\frac{3}{14}$ (C) $\frac{8}{21}$ (D) $\frac{6}{21}$ (E) $\frac{6}{10}$

17. $8.00 - 1.35 =$ (A) 6.75 (B) 7.65 (C) 7.35 (D) 6.5 (E) 6.65

18. $2.772 \div 7 =$ (A) .396 (B) .0025 (C) 3.96 (D) .296 (E) 0

19. $\frac{11}{12} \times 1\frac{1}{2} =$ (A) $1\frac{9}{24}$ (B) $1\frac{3}{8}$ (C) $\frac{11}{36}$ (D) $2\frac{1}{8}$ (E) $2\frac{3}{8}$

20. $.70 - .25 =$ (A) .55 (B) .45 (C) 5.5 (D) 4.5 (E) 5

21. $29.4 \times 1.4 =$ (A) 41.16 (B) 30.576 (C) 4.116 (D) 3.57 (E) 28

22. $\frac{9}{10} \div \frac{1}{5} =$ (A) $4\frac{1}{2}$ (B) $\frac{9}{2}$ (C) $5\frac{1}{2}$ (D) $7\frac{1}{15}$ (E) $\frac{9}{50}$

23. $.65 + 4.2 + 3.85 =$ (A) 8.7 (B) 8.52 (C) 8.2 (D) 14.55 (E) 8.8

24. $\frac{1}{4} \div 7 =$ (A) $\frac{7}{4}$ (B) $\frac{5}{21}$ (C) $7\frac{3}{4}$ (D) $7\frac{1}{4}$ (E) $\frac{1}{28}$

25. $.2 + .002 + 2 =$ (A) 22.2 (B) 22.02 (C) .22 (D) 2.202 (E) 2

26. $34.6 \times 15 =$ (A) 51.9 (B) 5.19 (C) 519 (D) 1076 (E) 4

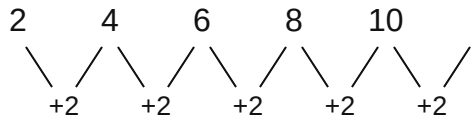
NUMERICAL REASONING

☞ **Directions:**

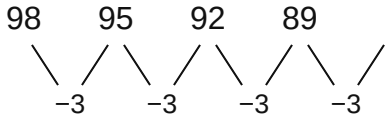
This test measures your ability to find a pattern in a series of numbers (a number series is a group of numbers given in a special order).

In each problem, a series of numbers is given which forms a specific pattern according to some rule. You are to study how the given numbers are related to each other, and then determine which number should come next in the series.

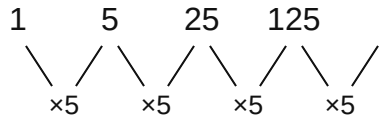
Now look at the following sample problems below. The relationship between the numbers is illustrated and the rule given in parentheses.



12 (Add 2 to each number in the series.)



86 (Subtract 3 from each number in the series.)



625 (Multiply each number in the series by 5.)

☞ TEST-TAKING TIPS: NUMERICAL REASONING

1. Read the series of numbers.

2. Look at the problem closely, and try to notice any patterns in the series. Estimate the approximate answer.

3. Try adding, subtracting, multiplying, or dividing numbers in the series from each other, beginning with those closest to each other in the series. Some patterns may require that you use two functions such as addition and multiplication (e.g. +2 x2 +2 x2).

4. Use the scratch paper for your calculations.

5. Compare any answer you get with your estimated answer.

6. Compare your answer with the answers given on the test.

    - If one of the answers on the test matches you answer, fill in the letter of the answer on the answer sheet.
    - If none of the answers on the test matches your answer, double check your calculations.

7. After you have finished all of the questions, if you have time, go back and double check your work.

✎ PRACTICE PROBLEMS: NUMERICAL REASONING

How to do a Numerical Reasoning problem on the test:

Step 1: Write the problem on your scratch paper.

Step 2: Try to recognize any simple patterns in the numbers.

Step 3: Try adding, subtracting, multiplying, or dividing numbers in the series.

Step 4: You should see a pattern in any numbers you add, subtract, multiply, or divide by in the series. Try to determine if there is a number or numbers common to all numbers in the series.

Step 5: If none of the answers on the test match your answer, recheck your arithmetic.

Practice Test 1:

____ 1.	5 10 20 40	A) 100	B) 60	C) 80	D) 120	E) 30
____ 2.	5 8 6 9 7 10	A) 8	B) 13	C) 7	D) 14	E) 23
____ 3.	74 70 66 62 58	A) 56	B) 64	C) 52	D) 50	E) 54
____ 4.	2 4 8 14 22	A) 44	B) 32	C) 30	D) 28	E) 34
____ 5.	243 81 27 9	A) 28	B) 4	C) 18	D) 3	E) 54
____ 6.	1 5 2 6 3 7	A) 21	B) 6	C) 4	D) 18	E) 10
____ 7.	1 4 16 64 256	A) 20	B) 48	C) 192	D) 1024	E) 16384

Answers for all practice problems can be found in the Appendix at the back of this manual.

Practice Test 2:

1.	**22**	**20**	**23**	**21**	**24**	**22**	**25**			
		A) 21		B) 23		C) 25		D) 27		E) 28
2.	**68**	**72**	**36**	**40**	**20**	**24**	**12**			
		A) 6		B) 16		C) 20		D) 24		E) 28

3. **16** **18** **21** **14** **16** **19** **12**

A) 5 B) 8 C) 9 D) 13 E) 14

4. **8** **24** **12** **36** **18** **54** **27**

A) 13 B) 18 C) 54 D) 69 E) 81

5. **24** **48** **12** **24** **6** **12** **3**

A) 6 B) 12 C) 18 D) 24 E) 48

6. **12** **10** **8** **16** **14** **12** **20**

A) 12 B) 14 C) 16 D) 18 E) 22

7. **13** **4** **14** **5** **15** **6** **16**

A) 6 B) 7 C) 8 D) 17 E) 18

8. **82** **73** **64** **55** **46** **37** **28**

A) 14 B) 18 C) 19 D) 20 E) 27

9. **78** **76** **38** **36** **18** **16** **8**

A) 3 B) 4 C) 5 D) 6 E) 7

10. **17** **19** **16** **20** **15** **21** **14**

A) 18 B) 19 C) 20 D) 21 E) 22

Problem Solving

Directions

This test measures your ability to determine an outcome, to record and retrieve information, to measure, and to use geometric concepts. Each problem represents a situation which you have probably encountered before.

In each problem, you will be presented with a detailed situation to respond to. You should read each problem carefully and decide what you are to do to respond to the situation. Look at the five possible answers given and determine which answer choice is best. Work out your answer on scratch paper, if needed. You are to mark the letter of the correct answer on the Answer Sheet. Make sure that your marks are heavy and black. Erase completely any answer you wish to change. Please do not make any marks in the test booklet.

Now try the following sample problems and mark the correct answers in the **Problem Solving Samples** box.

1. A roll of film has 24 exposures. Two will get ruined. How many good exposures will be left?

a. 24 *b.* 22 *c.* 20 *d.* 2 *e.* NG (not given)

2. How many squads are in a force of 200 people, if there are 20 people in each squad?

f. 20 *g.* 12 *h.* 5 *i.* 2 *k.* NG (not given)

The correct answer to the first sample problem is 22, since 24 minus 2 equals 22. Therefore, you should have blackened the b.

For the second sample problem, the correct answer is 10, since 200 divided by 20 equals 10. Therefore, you should have blackened the k because the correct answer was not given (NG).

Are there any questions about the sample problems?

You will have **35 minutes** to complete this test. Work as fast and accurately as possible.

OVER

Test-Taking Tips (Problem Solving)

Do you have any questions?

Study Suggestions:

1. Read the problem carefully. If you take two oranges from ten oranges, how many do you have?
2. Identify what you are asked to find.

   Larry needs to rope off a place for the work site, eight feet long and six feet wide, on the parking lot, ten feet from the flag pole. How much rope did he need? Here you are being asked to find how much rope will be required to rope off a work site.
3. Eliminate extraneous information. Determine what information you need to solve the problem.

   The fact that the work site must be ten feet from the flagpole is extraneous information. It is not necessary to solve the problem.
4. If possible, write a number sentence to determine what computation you must do.
5. Test your result against what you first decided to find. Does the answer make sense?
6. It is sometimes helpful to draw a picture of the problem to see what is known and what is unknown.

Example Problem: Eddie and Jose walked ten blocks before they realized they should have turned right at the corner three blocks before. So they retraced their steps to that corner, made the turn, and walked five blocks to the store they wanted. All together they walked 18 blocks. Which number sentence can you use to find how many unnecessary blocks they walked?

1. 3 X 2 =
2. 18 – (10 + 5) =
3. 7 + 5 =
4. 10 – 3 =

Drawing a picture would help to solve this problem. You also must focus on what needs to be found, the unnecessary blocks, not the necessary blocks. There also is extraneous information, such as the five blocks to the store. The correct answer is number 1.

Sample Problems:

1. An industrial plant produces 150 cars in 6 weeks. If the plant operates 5 days per week, how many cars does it produce per day?
2. A larger box contains 3 smaller boxes, and in each of these smaller boxes there are 4 tiny boxes. How many boxes are there altogether?
3. A file case has 21 drawers numbered from 1 to 21. The even numbered drawers contain 80 cards in each drawer. What is the number of cards in the even-numbered drawers?
4. An express train travels 3 miles in 2 minutes. What was the average rate in miles per hour?
5. An air conditioner which sold for $460 was on sale for a 50% discount. What is the sale price?

WORD PROBLEMS

One of the biggest problems encountered in shop math and algebra is in the attempt to put the information from a word problem into a workable formula. It is necessary to know the "code" of the word problem.

Up to now, solving problems has required adding, subtracting, multiplying, and dividing. The required information has been shown in signs. In word problems, the signs are no longer there, but have been replaced by words. A few of the words that are used are listed below.

ADD (+)

Sum of	More than or More	Increased by or Increase
Greater than	Plus	Added to
Combined with	In addition to	

SUBTRACT (–)

Less than	Diminished by	Decreased by
Minus	Subtracted	Difference
Remainder or remains	How many more	How much less
Reduced by	Left	Fewer than

MULTIPLY (×)

Product	Times	Multiplied by
Of	Twice or Three times, etc.	

DIVIDE (÷)

Quotient	Divide by or Divide equally	Split
Part of	Separate into parts	Shared

EQUAL (=)

Gives	Total	Result
Altogether	Is	

The statement "4 and 7" can be written mathematically by substituting the proper symbol for the word "and".

4 and 7

4 + 7

Whether the values are represented by numbers or letters has no effect on the substitution for the words representing the operation.

x diminished by 7

x – 7

Divisibility Rules:

Divisible by 2?

Rule: If it ends with a 0, 2, 4, 6, or 8

Divisible by 3?

Rule: If the sum of the digits is a multiple of 3. For example the sum of 2079 is 18 which is divisible by 3

Divisible by 4?

Rule: If the last two digits are a multiple of 4 (or if the last two digits are 00). For example we know that the number 4624 is divisible by 4 because 24 is a multiple of 4.

Divisible by 5?

Rule: If it ends with a 5 or a 0 it is divisible by 5.

Divisible by 6?

Rule: If it is divisible by both 2 and 3 it is divisible by 6. (Hint it has to be an even number). For example we know that 6294 is divisible by 3 because the sum of the digits is 21 and it is an even number.

Divisible by 7?

Rule: If you double the last digit and subtract it from the rest of the number and the answer is 0 or 7 the number is divisible by 7.

Example take 861 (double of 1 is 2, 86-2=84, and the double of 4 is 8, 8-8=0) (861/7=123)

Divisible by 8?

Rule: The number is divisible by 4 and the last three digits are divisible by 8.

Divisible by 9?

Rule: If the sum of the digits are a multiple of 9. For example the sum of 46926 is 27 which is a multiple of 9.

Divisible by 10?

Rule: If the last digit is a 0 it is divisible of 10

Divisible by 11?

Rule: If you sum every second digit and the answer is 0 or is divisible by 11.

Example take 1364 ((3+4) - (1+6) = 0) since the answer is 0 the number is divisible by 11

Example take 3729 ((7+9) - (3+2) = 11) the number is divisible by 11

Divisible by 12

Rule: The number is divisible by both 3 and 4. For example if we apply the rule for 3 and 4 from above we know that 648 is divisible by 12.

Link to divisibility rules: http://www.mathsisfun.com/divisibility-rules.html

FRACTIONS: HANDOUT #1

Write each of the following fractions as an equivalent with a denominator of 12

1. $\frac{5}{6}$ 2. $\frac{3}{4}$ 3. $\frac{5}{4}$ 4. $\frac{56}{84}$ 5. $\frac{143}{156}$

Reduce each fraction to the lowest terms and change improper fractions into a mixed numbers.

6. $\frac{36}{20}$ 7. $\frac{14}{98}$ 8. $\frac{105}{30}$ 9. $\frac{66}{84}$ 10. $\frac{102}{114}$

Multiply and reduce to lowest terms

11. $\frac{3}{5} \times \frac{4}{7}$

12. $\frac{3}{4} \times 9$

13. $\frac{2}{3} \times \frac{4}{5} \times \frac{1}{3}$

14. $\frac{196}{27} \times \frac{9}{392}$

15. $\frac{3}{5} \times 15 \times \frac{8}{9}$

16. $\frac{72}{35} \times \frac{55}{108} \times \frac{7}{110}$

17. $5\frac{1}{8} \times 2\frac{2}{3}$

18. $3\frac{1}{2} \times 2\frac{1}{6}$

19. $2\frac{1}{2} \times 3\frac{1}{3} \times 1\frac{1}{2}$

20. $\frac{7}{8} \times 6 \times 1\frac{5}{6}$

Divide and reduce to lowest terms,

21. $\frac{1}{3} \div 2$

22. $8 \div \frac{3}{4}$

23. $\frac{3}{5} \div 2$

24. $\frac{25}{46} \div \frac{40}{69}$

25. $\frac{28}{125} \div \frac{5}{2}$

26. $\frac{30}{27} \div 6$

27. $1\frac{4}{5} \div 2\frac{5}{6}$

28. $12 \div 3\frac{1}{6}$

29. $12\frac{6}{7} \div 3$

30. $\frac{7}{8} \div (1\frac{1}{4} \div 4)$

31. The jaws of a vise will close a distance of $\frac{3}{8}$" for each turn of the handle. How far will the jaws close in only one half turn?

32. Find the total voltage supplied by three, $1\frac{1}{2}$ V dry cells connected in series.

33. An electrician gets time and one half for overtime. If he worked $4\frac{3}{4}$ hours overtime, how many hours will he be paid for this overtime?

34. If friction, copper and iron lose one twentieth of the energy put into a motor, how many kilowatts are lost if the motor receives $\frac{2}{5}$ KW?

35. About $\frac{2}{5}$ of the electrolyte in a storage battery is acid. If the battery contains $\frac{11}{16}$ lbs. of electrolyte, how much acid is in the cell of the battery?

36. What length of BX cable is needed to obtain six pieces, each $4\frac{1}{4}$' long?

37. How many pieces of pipe that are $\frac{2}{3}$' long must be laid together to make a pipe 16' long?

38. A bag of screws holds exactly $\frac{1}{4}$ pound. How many of these bags can be filled from $12\frac{2}{3}$ pound of screws?

39. If 1 gallon of gas costs 159 $\frac{9}{100}$ cents, how much does $\frac{1}{2}$ gallon cost?

40. What is the quotient (division) of $\frac{3}{8}$ and $\frac{5}{8}$?

FRACTIONS: HANDOUT #2

Add or subtract and reduce to lowest terms.

1. $\frac{3}{6} + \frac{1}{6}$
2. $\frac{2}{5} + \frac{3}{5}$
3. $\frac{5}{8} - \frac{3}{8}$
4. $\frac{6}{7} - \frac{1}{7}$
5. $\frac{1}{4} + \frac{2}{4} + \frac{3}{4}$
6. $\frac{2}{5} + \frac{3}{5} + \frac{4}{5}$
7. $\frac{7}{9} - \frac{4}{9}$
8. $\frac{9}{8} - \frac{1}{8}$
9. $\frac{3}{20} + \frac{1}{20} + \frac{4}{20}$
10. $\frac{5}{4} + \frac{4}{4} + \frac{3}{4}$

Find the LCD for each of the following, then add or subtract and reduce to lowest terms.

11. $\frac{1}{8} + \frac{3}{4}$
12. $\frac{5}{6} - \frac{1}{12}$
13. $2 + \frac{1}{3}$
14. $2 - \frac{3}{4}$
15. $\frac{2}{3} + \frac{3}{4}$
16. $\frac{1}{2} - \frac{1}{5}$
17. $\frac{5}{12} + \frac{3}{8}$
18. $\frac{9}{16} + \frac{7}{12}$
19. $\frac{8}{30} - \frac{1}{20}$
20. $\frac{9}{40} - \frac{1}{30}$

Add or subtract the following mixed numbers and reduce to lowest terms.

21. $4\frac{3}{10} + 8\frac{1}{10}$
22. $12\frac{5}{12} - 7\frac{1}{12}$
23. $1\frac{5}{8} + 2\frac{1}{2}$
24. $6\frac{1}{3} - 4\frac{1}{4}$
25. $6\frac{1}{2} + 2\frac{5}{14}$
26. $5\frac{4}{10} - 3\frac{1}{3}$
27. $1\frac{5}{6} + 2\frac{3}{4} + 5\frac{1}{2}$
28. $9\frac{4}{7} - 7\frac{2}{3}$
29. $18\frac{7}{12} + 19\frac{3}{16} + 10\frac{2}{3}$
30. $21\frac{2}{5} - 20\frac{5}{16}$

31. What is the total horsepower delivered by a $\frac{1}{3}$, a $\frac{1}{4}$ and a $\frac{1}{8}$ HP motor?

32. What length of bolt is covered by a lock washer ($\frac{1}{32}$ "), a washer ($\frac{1}{8}$") and a nut ($\frac{5}{16}$)?

33. The length of threading on a BX box is $\frac{7}{16}$ ". If $\frac{7}{32}$ " of thread is covered with washers, locknuts and metal, how much thread is left to attach a bushing?

34. An electrician had a $16\frac{1}{4}$ ft, long piece of conduit. If she used 11 $\frac{5}{12}$ ', how many feet of conduit were left?

35. A carbon brush, $\frac{15}{16}$ " in thickness, has a copper coating of $\frac{1}{64}$ " thickness on each side. What is the thickness of the carbon?

36. What is the total weight on three coils of wire weighing $\frac{16}{14}$, $4\frac{1}{8}$ and $2\frac{1}{2}$ pounds?

37. How much conduit is needed to pipe five receptacles, if the distance between them is: $6\frac{1}{4}$', 8 $\frac{7}{12}$ ', $10\frac{5}{8}$' and 4 $\frac{5}{10}$ '?